Phuong-Linh Huynh-Ha

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Research Experience

- **KIT Computer Vision Lab – Kyoto Institute of Technology (KIT)** Jan. 2026 – Mar. 2026
  Research Intern, supervised by Prof. Nobuhara Shohei
  - Awarded the **JASSO Scholarship by the Japanese government** and selected for the **KIT Global Internship Program** based on academic excellence.
  - Developing a **3D reconstruction framework** for **near-rotational motion** and **low-parallax environments**. Resolved motion ambiguity in **feature-poor scenes** by integrating **2D correspondences** to rectify and enhance SLAM-based estimations.
- **Software Engineering Laboratory – SELab HCMUS** Aug. 2025 – Present
  Thesis Student - Supervised by Assoc. Prof Minh-Triet Tran on Computer Vision
  - Co-authored and published "**Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**" at SOICT 2025 (Springer CCIS).
  - The thesis focuses on developing a method called **"EndoFGS"** that works on the **frequency domain** for **surgical scene reconstruction** designed to overcome challenges in feature-poor environments, such as blood vessels and tissue surfaces.
  - Participating in the **CVPR 2026 Spiideo SoccerNet SynLoc Challenge**; developing methods to detect and localize athletes in single-frame world coordinates utilizing synthetic data.

Selected Publications

- Hoang-Phuc Nguyen, Phuong-Linh Huynh-Ha, Minh-Triet Tran. **Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**. *SOICT 2025, Springer CCIS.*
  Introduced a **training-free, anchor-guided geometric mapping approach** using Delaunay triangulation to resolve overfitting and improve fine-grained keypoint accuracy in category-agnostic pose estimation.

Education

- **University of Science (HCMUS), Ho Chi Minh City, Vietnam** Sep. 2022 – Present
  Advanced Program in Computer Science; GPA: 3.9/4.0
  - Granted **Merit-Based Scholarship (Top 4 GPA)** in Semester 2, 2024-2025.
  - Being 1 of **5 recipients** of **Career Readiness Scholarship**, awarded by **Gameloft** in partnership with **HCMUS** for demonstrating strong professional potential and academic merit.
  - Achieved **Top 1.9% score on the VNU-HCMC Assessment Test** (HSA)
  - Relevant coursework: Artificial Intelligence, Computer Vision, Natural Language Processing, Information Retrieval, Computer Systems, Statistics.

Achievements and Awards

- **Artificial Intelligence Olympiad for University Students (3-person team)** Nov. 2025 – Dec. 2025
  Organized by Vietnam Association for Information Processing (VAIP)
  - Awarded **Third Prize** at the **National Finals** (Dec. 2025).
  - Awarded **First Prize**, **Champion** across **all Regional Preliminary Rounds** (Nov. 2025).
- **Odon Vallet Scholarship** Aug. 2022
  - Awarded prestigious **Odon Vallet Fellowship** for academic excellence, issued by Rencontres du Vietnam.
- **Vietnam National Olympiad in English** Apr. 2022, Feb. 2021
  - **Bronze Medal** in 2022 and **Honorable Mention** in 2021, awarded by Vietnam Ministry of Education and Training.
  - **First Place** in the Provincial Olympiad in English of Quang Ngai throughout 2021 - 2022.

Projects

- **Robust 3D Reconstruction via 2D Correspondence Integration** — Researching
  - Extracting and comparing optical flow from MegaSaM and CoTracker3 to resolve motion ambiguity in low-parallax and near-rotational videos.
  - Using heatmap analysis to visualize flow differences, creating a reliable method to correct SLAM tracking errors in feature-poor environments.
- **EndoFGS: Surgical Scene Reconstruction & 3D Gaussian Splatting** — Researching
  - Building a surgical scene reconstruction pipeline using 3D Gaussian Splatting, evaluated on the EndoNeRF and StereoMIS datasets.
  - Improving rendering quality (PSNR/SSIM) on textureless tissues by applying frequency-domain filters, including Wavelet, Fourier, and high-boost, to enhance image features.
- **SoccerNet SynLoc: Athlete Detection & Localization** — Researching
  - Developing a monocular athlete localization pipeline for the CVPR 2026 challenge using YOLO26 for bounding box detection and RTMPose for pelvis keypoint extraction.
  - Mapping 2D pixel coordinates to 3D stadium coordinates by applying pitch-based camera calibration and ground projection techniques.

Extracurricular Activities

- **Youth Union Branch Commendation (Youth Union Branch 22APCS2)** — 2022–2023
  Committee Member – University of Science, VNU-HCM
  - Received a **Certificate of Merit** from the Principal for achieving good results in Youth Union organizational work and youth movements during the 2022-2023 academic year.
- **Inspire to Aspire Seminar Series** — 2023
  Organizer – Ho Chi Minh City, Vietnam
  - Organized a seminar series titled *"Inspire to Aspire"* that connected students with experienced professionals in technology and business, providing career insights and fostering industry awareness.
- **Asian-Pacific Children's Convention (APCC)** — Jul. 2015
  Youth Delegate – Fukuoka, Japan
  - Selected by the Ho Chi Minh Communist Youth Union to represent Vietnam at APCC in Fukuoka, Japan, focusing on international cultural exchange to foster global understanding, peace, and friendship.

Skills & Interests

Technical Skills:	Scientific Research, Problem Solving, Machine Learning, Computer Vision
Technical Languages:	C/C++, Python, Kotlin, SQL, LaTeX, Markdown, CSS, HTML
Frameworks/Libraries:	PyTorch, NumPy, Pandas, R, Jetpack Compose, SFML
Tools:	Google Colab, Docker, Git/GitHub, Android SDK, 3D Slicer
Languages:	English (IELTS 8.0), Vietnamese (native), Japanese (for daily conversation)
Interests:	Cycling, Swimming, Singing, Guitar (learning), Poker